B.E. Even Semester Midterm Examination, 2024-25

Subject: Economics and Accountancy | **Paper Code:** HSM HU 401 | **Full Marks:** 15

Q1. What do you mean by engineering economics? Why engineering economics is important?

Answer: Engineering Economics is the application of economic techniques to evaluate design alternatives. It bridges engineering and economics to quantify costs/benefits of technical projects. Important for: Decision Making, Cost Analysis (life-cycle costing), Resource Optimization, and Feasibility Studies.

Q2. Discuss different types of market with examples.

Answer: 1. **Perfect Competition:** Many buyers/sellers, homogeneous products. (e.g., Agriculture) 2. **Monopoly:** Single seller, unique product, high barriers. (e.g., Indian Railways) 3. **Monopolistic Competition:** Many sellers, differentiated products. (e.g., Restaurants) 4. **Oligopoly:** Few large firms, interdependent. (e.g., Telecom, Airlines)

Q3. Write short notes on GDP.

Answer: Gross Domestic Product (GDP) is the total monetary value of all finished goods and services produced within a country's borders in a specific time period. Formula: $GDP = C + I + G + (X - M)$. It's a scorecard of a country's economic health.

Q4. Discuss major exception of law of demand.

Answer: Exceptions occur where demand rises when price increases: 1. **Giffen Goods:** Highly inferior staple foods where a price increase forces people to buy more as they can't afford substitutes. 2. **Veblen Goods:** Luxury items (diamonds) bought as status symbols. 3. **Expectation of Future Prices:** Buying more now before prices rise further.

Q5. Briefly discuss 'returns to scale'.

Answer: Describes how output changes when there's a proportionate change in all inputs in the long run: 1. Increasing Returns: Output increases by a greater proportion than inputs. 2. Constant Returns: Output increases in exact proportion to inputs. 3. Decreasing Returns: Output increases by a smaller proportion than inputs.